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Lab 10

TASK 1

1. ▾ Assets 4

winpmem_mini_x64_rc2.exe	515 KB	Oct 13, 2020

Today (1)

winpmem_mini_x64_rc2

Downloaded winpmem_mini_x64_rc2.exe.

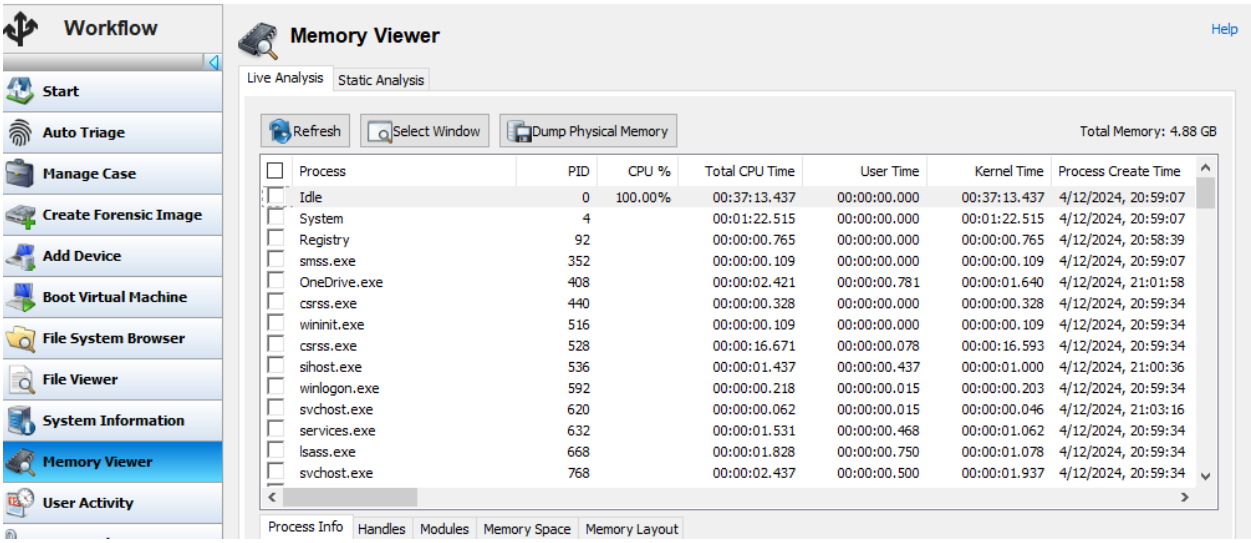
2. Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd c:\users\navid\Downloads

c:\Users\navid\Downloads>
```

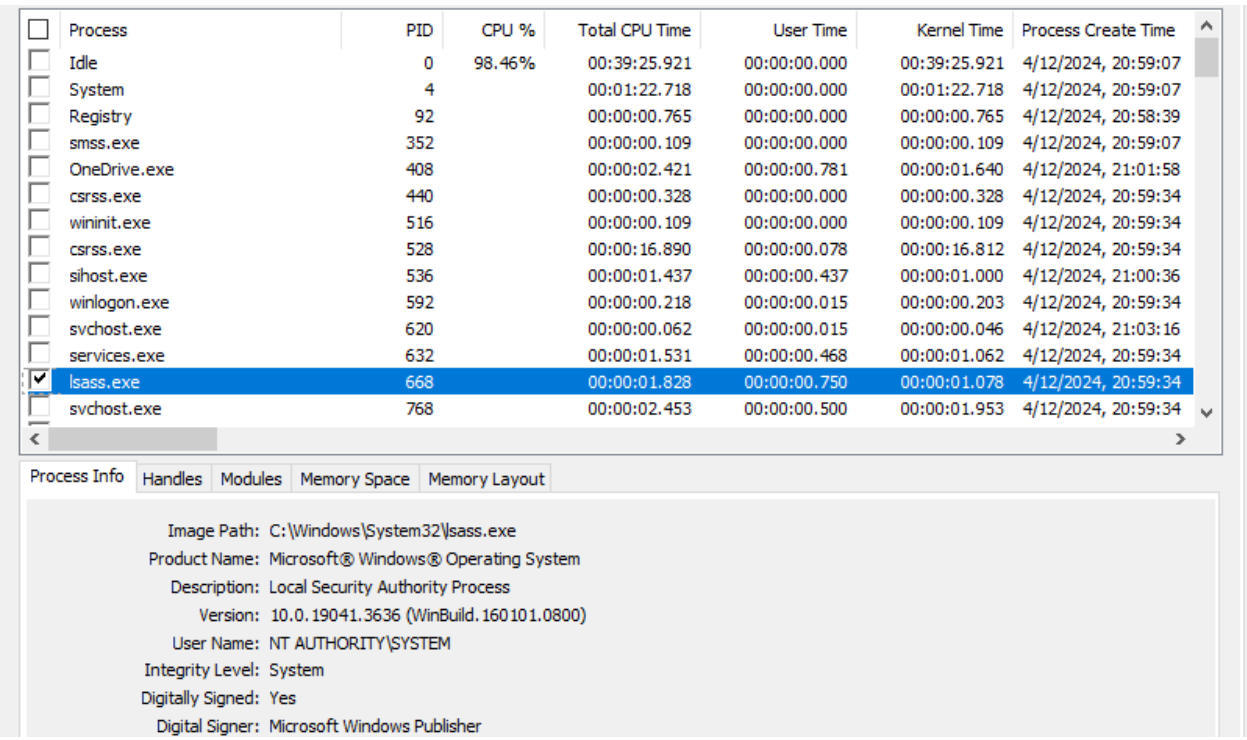
Launched CMD as administrator and switched directories to downloads.

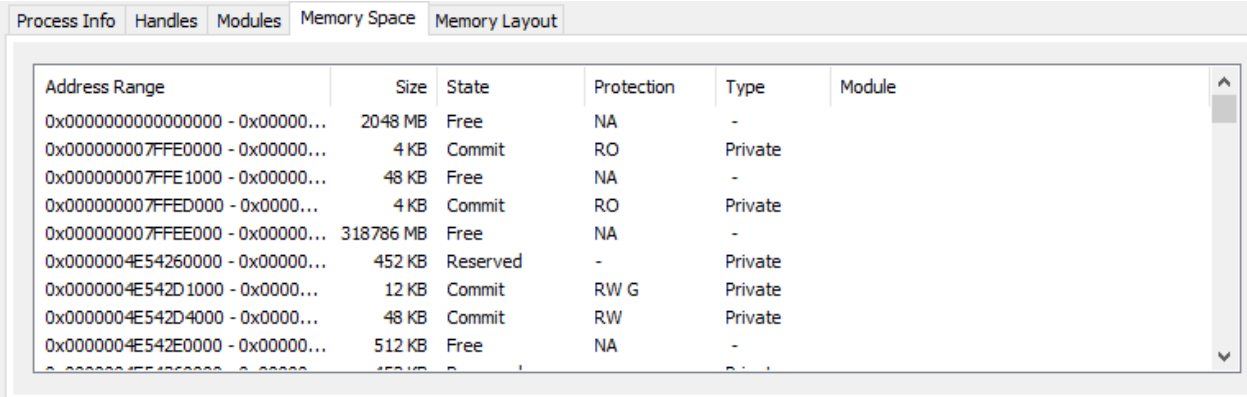
```
c:\Users\navid\Downloads>winpmem_mini_x64_rc2.exe mem.raw
WinPmem64
Extracting driver to C:\Users\navid\AppData\Local\Temp\pmeAEFA.tmp
Driver Unloaded.
Loaded Driver C:\Users\navid\AppData\Local\Temp\pmeAEFA.tmp.
Deleting C:\Users\navid\AppData\Local\Temp\pmeAEFA.tmp
The system time is: 04:07:03
Will generate a RAW image
 - buffer_size_: 0x1000
CR3: 0x00001AA000
 4 memory ranges:
Start 0x00001000 - Length 0x0009E000
Start 0x00100000 - Length 0x00002000
Start 0x00103000 - Length 0xDFEED000
Start 0x100000000 - Length 0x58A00000
max_physical_memory_ 0x158a00000
Acquitision mode PTE Remapping
Padding from 0x00000000 to 0x00001000
pad
 - length: 0x1000
```

Ran winpmem_mini_x64_rc2.exe mem.raw. This runs winpmem and outputs to a mem.raw file.

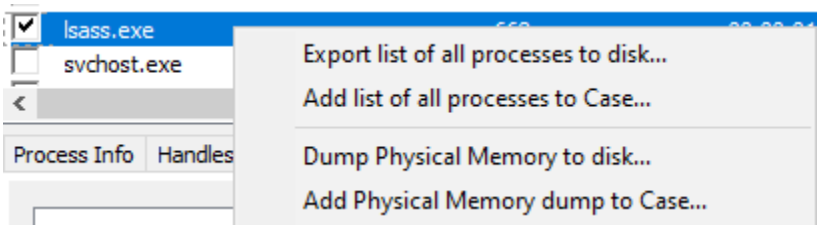
3. 

Launched OSForensics and selected the Memory Viewer tab.

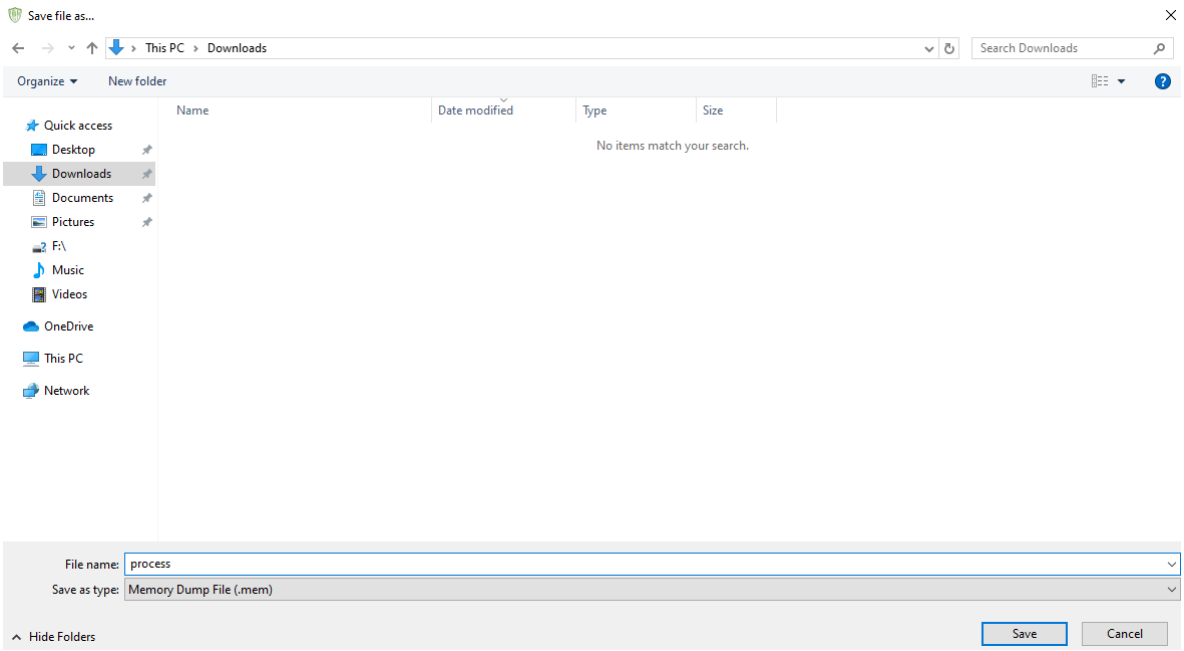


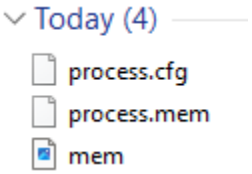


Selected lsass.exe and viewed the process info and memory space.



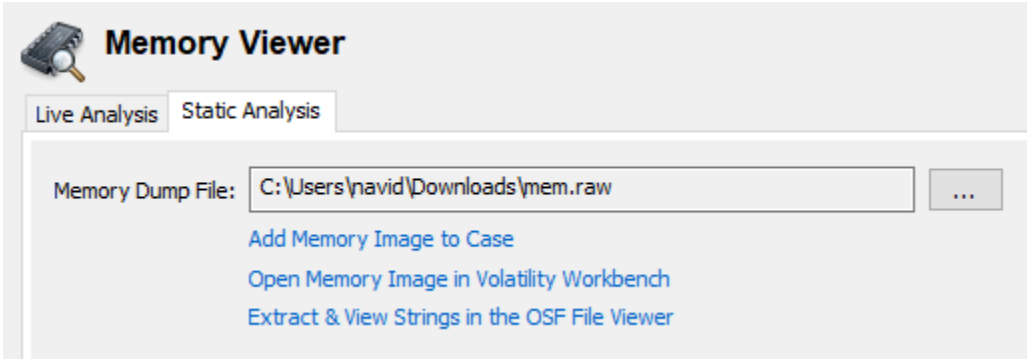
Right clicked the process and selected dump physical memory to disk.



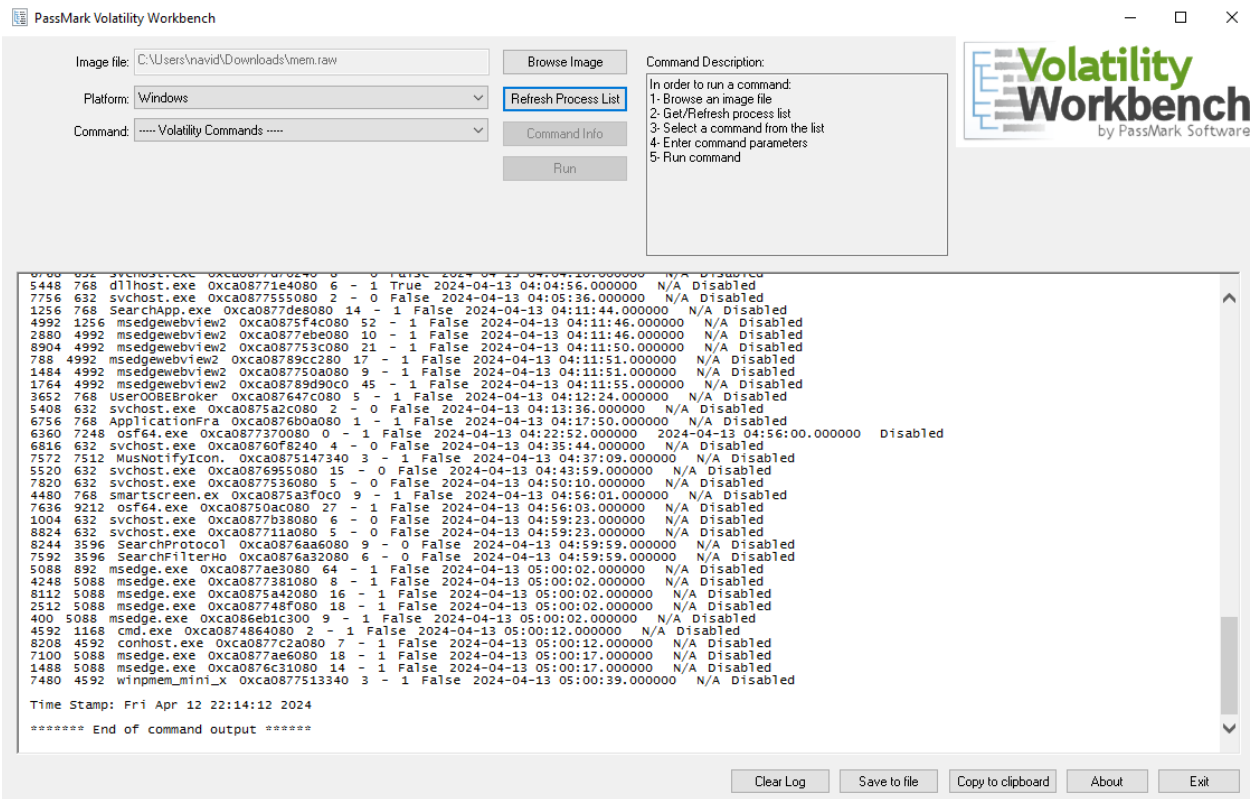


Named the file “process” and saved it to the downloads folder.

4.



Navigated to the Static Analysis tab in Memory Viewer and set the memory dump file as mem.raw. Selected “Open Memory Image in Volatility Workbench”.

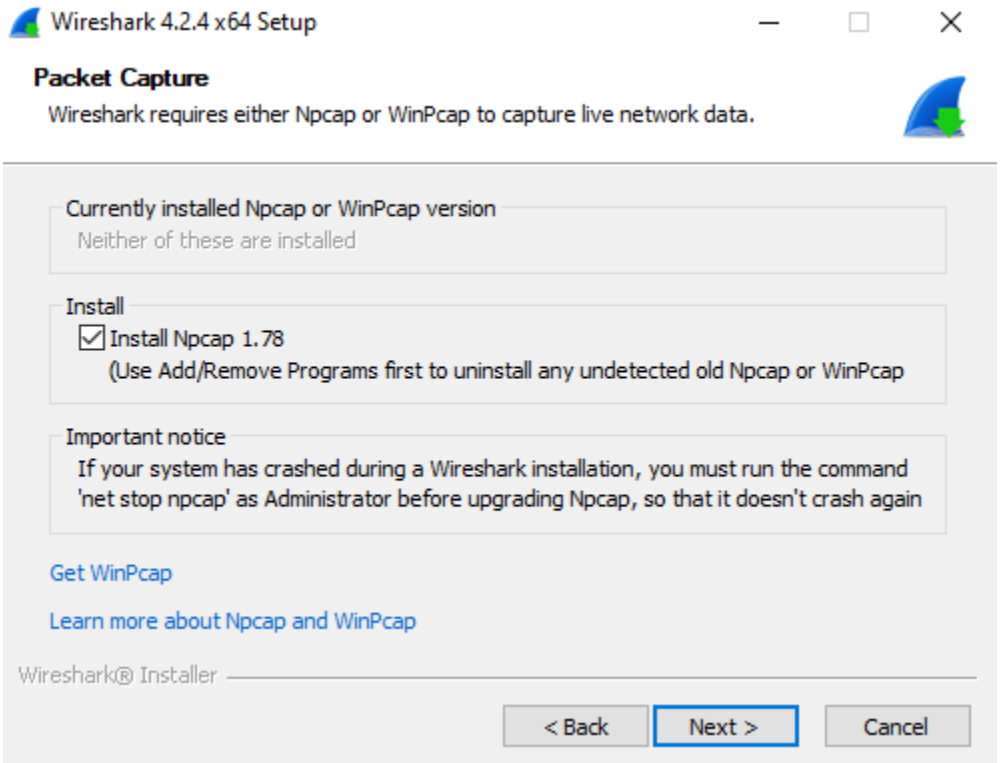


In Volatility Workbench, I pressed “Get Process List”, and it listed all the processes that were running when the snapshot was taken.

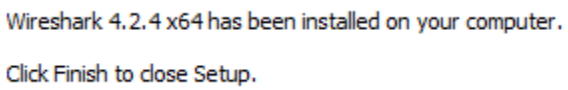
TASK 2

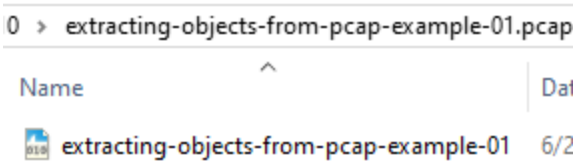
1.

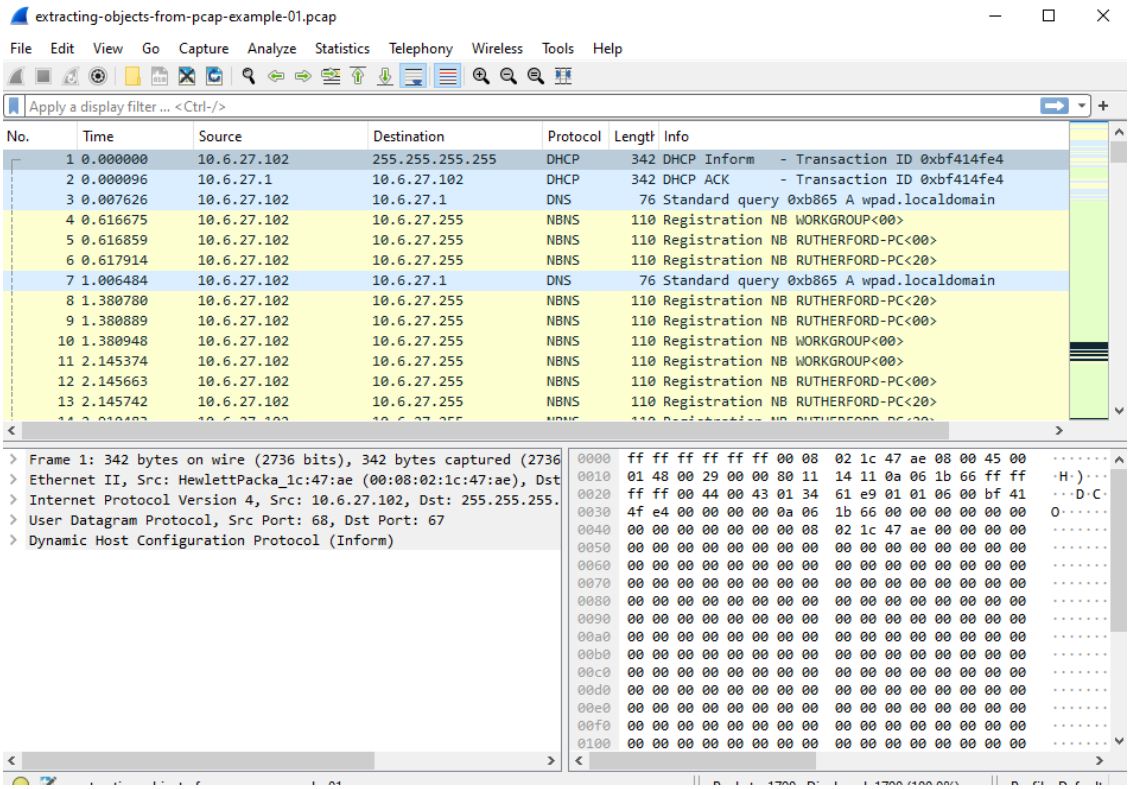
Downloaded Wireshark Windows x64 Installer.



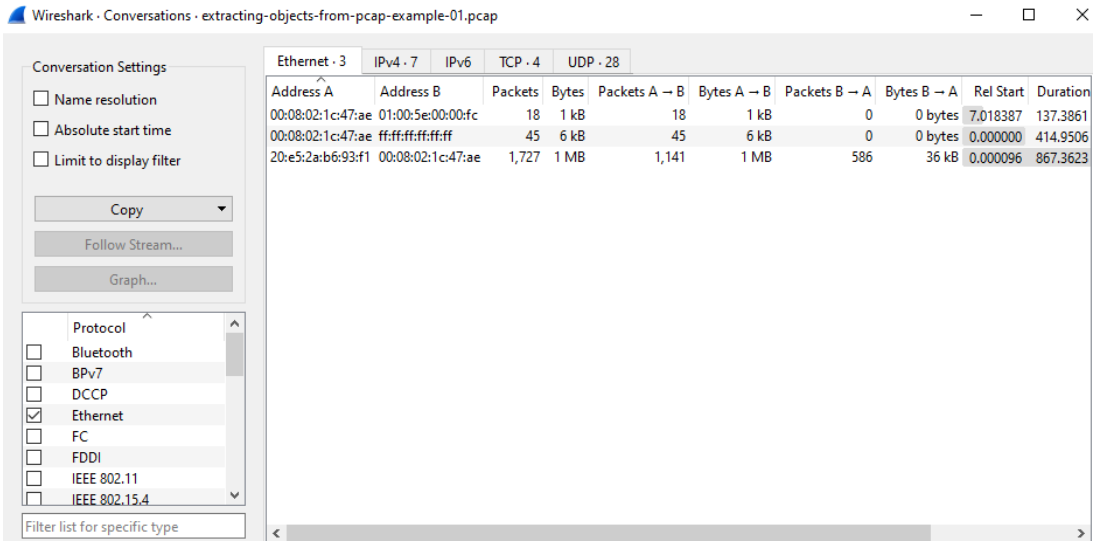
Followed the installation wizard and installed the program. Made sure Npcap was installed as well.



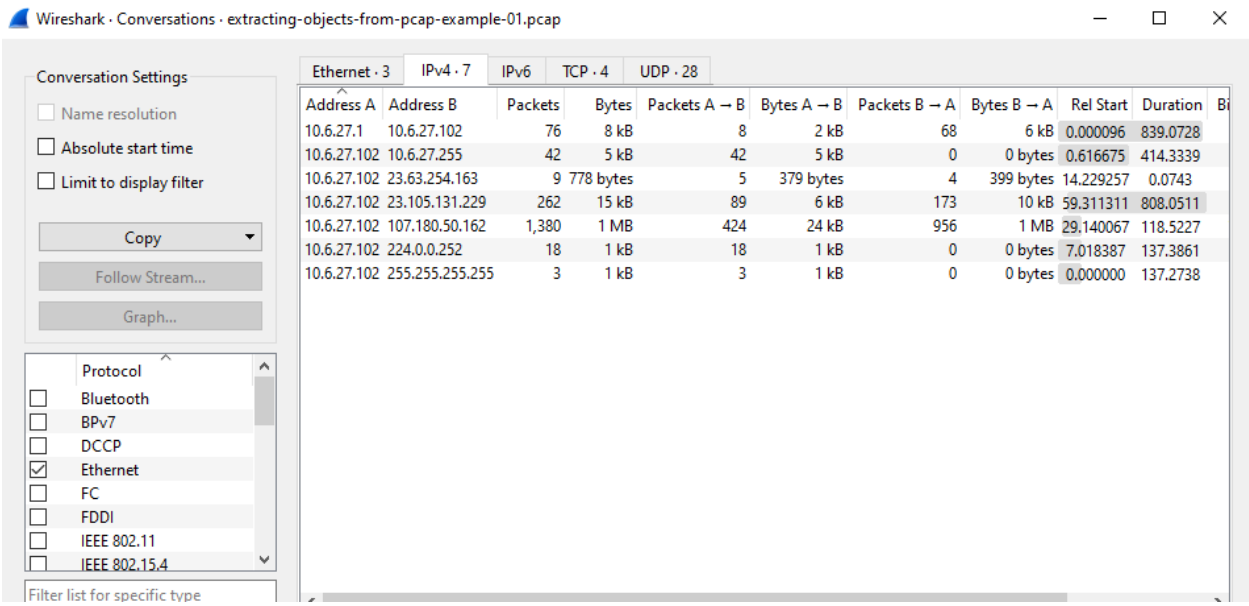
2. 



Extracted the zip file and double clicked it to open it in Wireshark.

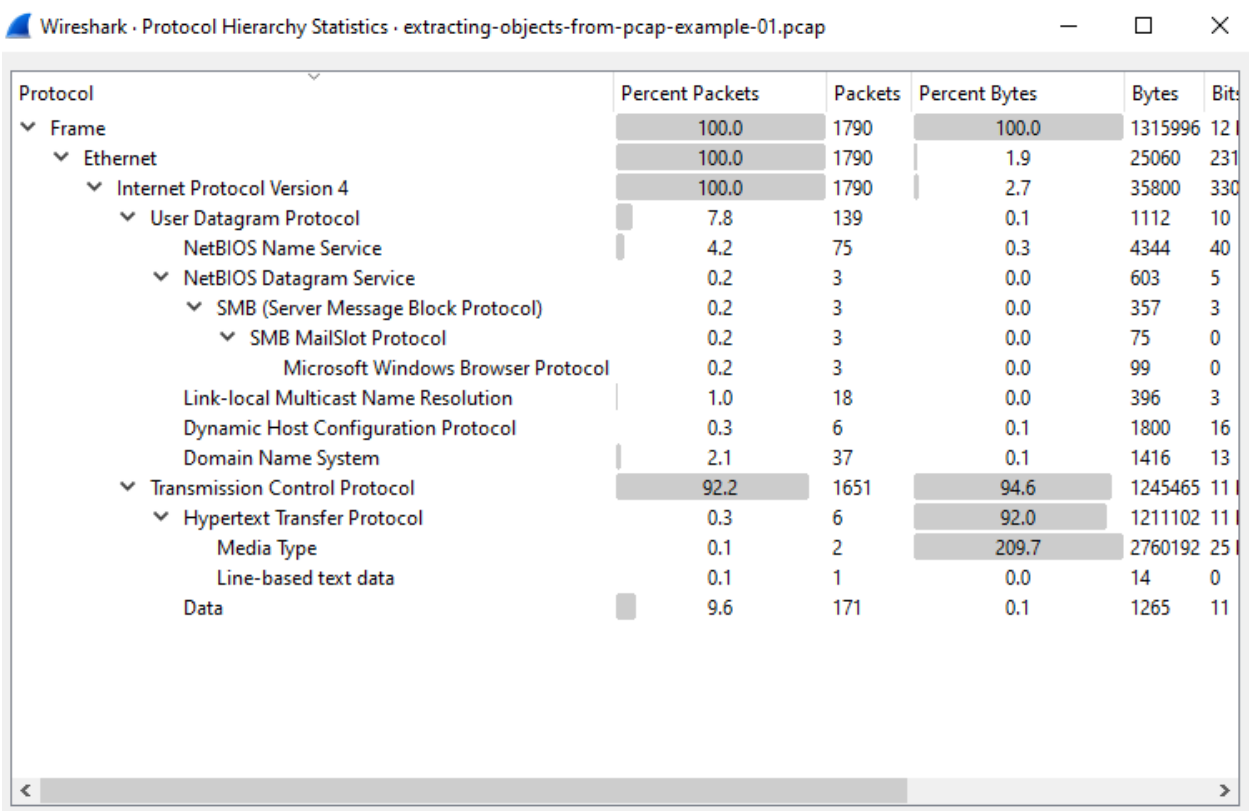
3. 

Selected “Conversations” under the Statistics menu.



Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration
10.6.27.1	10.6.27.102	76	8 kB	8	2 kB	68	6 kB	0.000096	839.0728
10.6.27.102	10.6.27.255	42	5 kB	42	5 kB	0	0 bytes	0.616675	414.3339
10.6.27.102	23.63.254.163	9	778 bytes	5	379 bytes	4	399 bytes	14.229257	0.0743
10.6.27.102	23.105.131.229	262	15 kB	89	6 kB	173	10 kB	59.311311	808.0511
10.6.27.102	107.180.50.162	1,380	1 MB	424	24 kB	956	1 MB	29.140067	118.5227
10.6.27.102	224.0.0.252	18	1 kB	18	1 kB	0	0 bytes	7.018387	137.3861
10.6.27.102	255.255.255.255	3	1 kB	3	1 kB	0	0 bytes	0.000000	137.2738

Switching to the IPv4 tab shows some addresses, packet counts, byte sizes, etc.



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits
Frame	100.0	1790	100.0	1315996	12
Ethernet	100.0	1790	1.9	25060	231
Internet Protocol Version 4	100.0	1790	2.7	35800	330
User Datagram Protocol	7.8	139	0.1	1112	10
NetBIOS Name Service	4.2	75	0.3	4344	40
NetBIOS Datagram Service	0.2	3	0.0	603	5
SMB (Server Message Block Protocol)	0.2	3	0.0	357	3
SMB MailSlot Protocol	0.2	3	0.0	75	0
Microsoft Windows Browser Protocol	0.2	3	0.0	99	0
Link-local Multicast Name Resolution	1.0	18	0.0	396	3
Dynamic Host Configuration Protocol	0.3	6	0.1	1800	16
Domain Name System	2.1	37	0.1	1416	13
Transmission Control Protocol	92.2	1651	94.6	1245465	11
Hypertext Transfer Protocol	0.3	6	92.0	1211102	11
Media Type	0.1	2	209.7	2760192	25
Line-based text data	0.1	1	0.0	14	0
Data	9.6	171	0.1	1265	11

Opened the Protocol Hierarchy from the Statistics tab.

Hypertext Transfer Protocol	0.3	6	92.0	1211102	11 k	3
Media Type	0.1	2	209.7	2760192	25 k	2
Line-based text data	0.1	1	0.0	14	0	1
Data	9.6	171	0.1	1265	11	171

The summary includes HTTP traffic.

http

No.	Time	Source	Destination	Protocol	Length	Info
43	14.272449	10.6.27.102	23.63.254.163	HTTP	151	GET /ncsi.txt HTTP/1.1
45	14.302997	23.63.254.163	10.6.27.102	HTTP	233	HTTP/1.1 200 OK (text/plain)
71	29.202755	10.6.27.102	107.180.50.162	HTTP	343	GET /Documents/Invoice&MSO-Request.doc HTTP/1.1
337	33.648846	107.180.50.162	10.6.27.102	HTTP	162	HTTP/1.1 200 OK (application/msword)
356	38.470797	10.6.27.102	107.180.50.162	HTTP	361	GET /knr.exe HTTP/1.1
1456	39.117888	107.180.50.162	10.6.27.102	HTTP	243	HTTP/1.1 200 OK (application/x-msdownload)

On the main page, I filtered the results to HTTP only. It seems the user has downloaded 3 things, a txt file, a doc file, and an exe file.

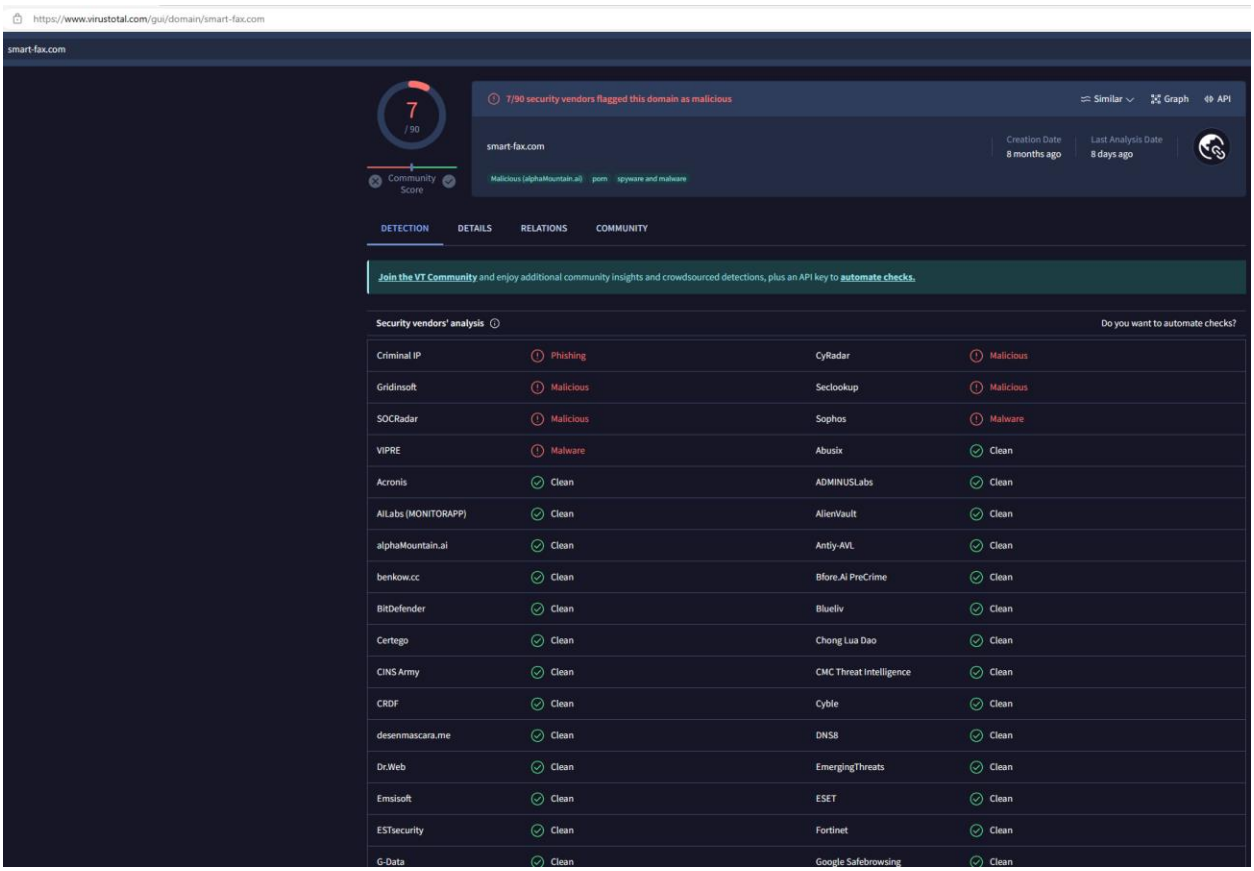
View Go Capture Analyze Statistics Telephony Wireless Tools Help

- ✓ Main Toolbar
- ✓ Filter Toolbar
- ✓ Status Bar
- Full Screen F11
- ✓ Packet List
- ✓ Packet Details
- ✓ Packet Bytes
- Packet Diagram
- Time Display Format
- Name Resolution
  - Edit Resolved Name
  - ✓ Resolve Physical Addresses
  - Resolve Network Addresses
  - Resolve Transport Addresses
- Zoom
- Expand Subtrees Shift+Right
- Collapse Subtrees Shift+Left

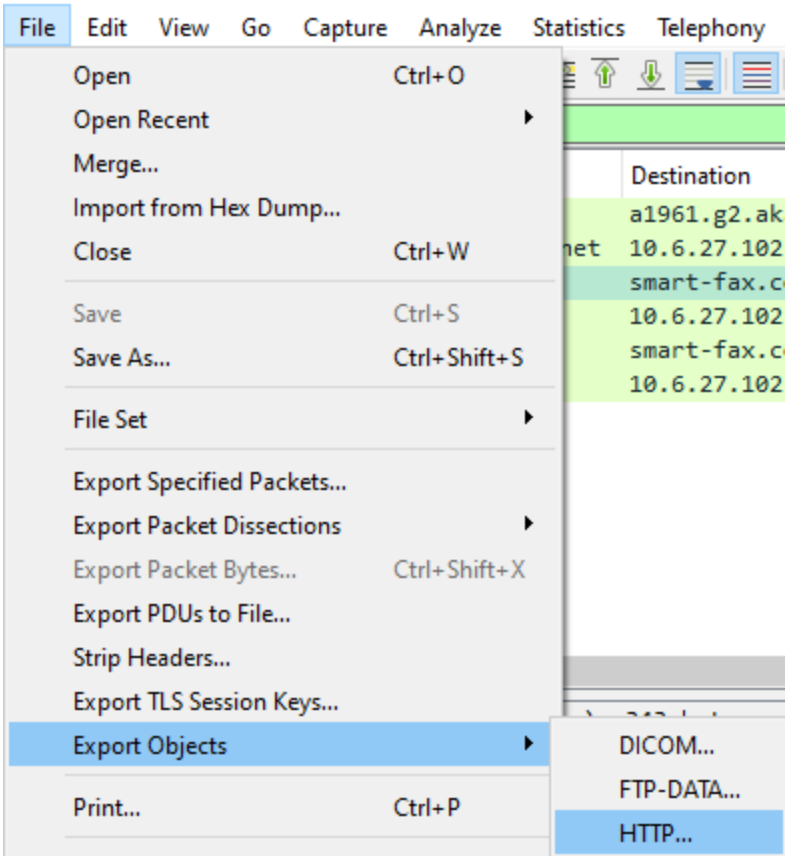
Switched the Name Resolution to Resolve Network Addresses

No.	Time	Source	Destination	Protocol	Length	Info
43	14.272449	10.6.27.102	a1961.g2.akamai.net	HTTP	151	GET /ncsi.txt HTTP/1.1
45	14.302997	a1961.g2.akamai.net	10.6.27.102	HTTP	233	HTTP/1.1 200 OK (text/plain)
71	29.202755	10.6.27.102	smart-fax.com	HTTP	343	GET /Documents/Invoice&MSO-Request.doc HTTP/1.1
337	33.648846	smart-fax.com	10.6.27.102	HTTP	162	HTTP/1.1 200 OK (application/msword)
356	38.470797	10.6.27.102	smart-fax.com	HTTP	361	GET /knr.exe HTTP/1.1
1456	39.117888	smart-fax.com	10.6.27.102	HTTP	243	HTTP/1.1 200 OK (application/x-msdownload)

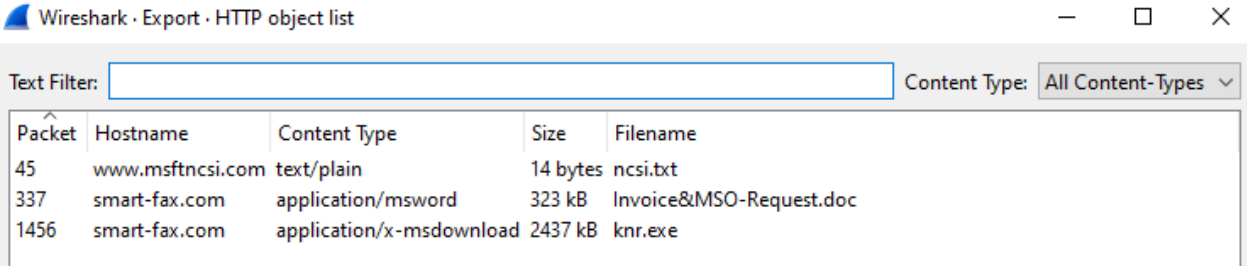
One of the addresses now shows as “smart-fax.com”. It seems that is where the user downloaded 2 of the files from.



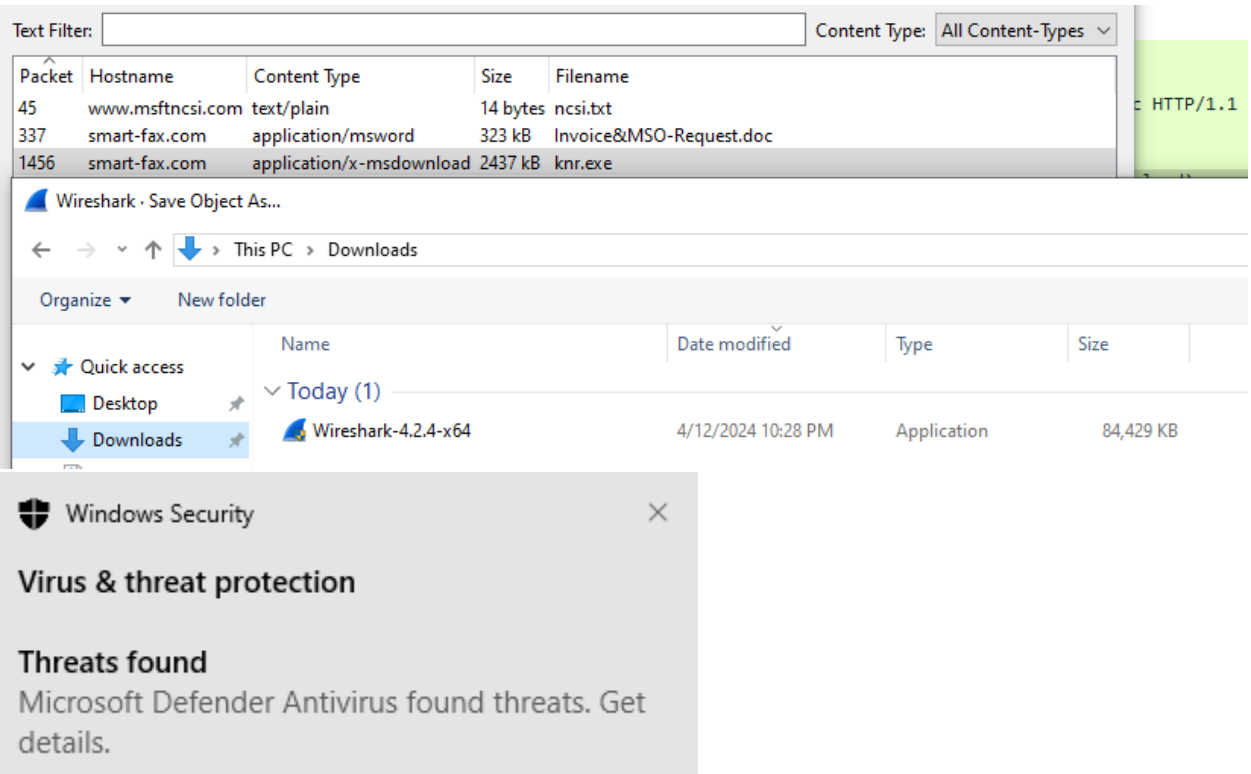
Searching “smart-fax.com” on virustotal.com shows that 7/90 vendors have flagged the domain as malicious.



Back on Wireshark, I exported the HTTP objects.



Here are the three downloaded files. The txt, doc, and exe. There's a good chance the exe is malware.



Saving the exe file to my VM resulted in this Windows Security notification. It detected a threat and automatically deleted it from my downloads folder.